CHAPTER 14 · DERIVATIONS

CAPACITORS

5 derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 CAPACITANCE OF A PARALLEL PLATE CAPACITOR

SYLLABUS 14.1(D)

RESULT $C_0 = \dfrac{\varepsilon_0 A}{d}$

1. Plates of area A, separation d, charges ±Q: $\sigma = \dfrac{Q}{A}$.
2. Gauss's law: the field between the plates is $E = \dfrac{\sigma}{\varepsilon_0} = \dfrac{Q}{\varepsilon_0 A}$ (uniform).
3. $V = Ed = Qd/(\varepsilon_0 A)$.
4. $\mathbf{C_0 = \dfrac{Q}{V} = \dfrac{\varepsilon_0 A}{d}}$.

2 CAPACITORS IN SERIES AND IN PARALLEL

SYLLABUS 14.3(A)

RESULT series: $\dfrac{1}{C} = \dfrac{1}{C_1} + \dfrac{1}{C_2} + \ldots$; parallel: $C = C_1 + C_2 + \ldots$

1. **Series**: each capacitor carries the same charge Q, and $V = V_1 + V_2 + V_3$.
2. $\dfrac{Q}{C} = \dfrac{Q}{C_1} + \dfrac{Q}{C_2} + \dfrac{Q}{C_3}$, so $\mathbf{\dfrac{1}{C} = \dfrac{1}{C_1} + \dfrac{1}{C_2} + \dfrac{1}{C_3}}$.
3. **Parallel**: each capacitor has the same p.d. V, and $Q = Q_1 + Q_2 + Q_3$.
4. $CV = C_1V + C_2V + C_3V$, so $\mathbf{C = C_1 + C_2 + C_3}$.

Energy formulae $U = \frac{1}{2}QV = \frac{1}{2}CV^2 = \dfrac{Q^2}{2C}$: derivation not required (14.4).

3 WHY A DIELECTRIC INCREASES THE CAPACITANCE (QUALITATIVE)

SYLLABUS 14.3

RESULT $C = \dfrac{Q}{V}$ increases

1. When the dielectric is inserted, its positive nuclei are attracted towards the negative plate and its electrons towards the positive plate.
2. Dipoles are formed (the dielectric is polarised); the induced charges set up an electric field opposite to the applied field.
3. The net field between the plates decreases: the positive plate becomes less positive and the negative plate less negative in potential.
4. The potential difference V between the plates decreases while the charge Q remains constant (isolated capacitor).
5. From $C = \dfrac{Q}{V}$, the **capacitance increases**.

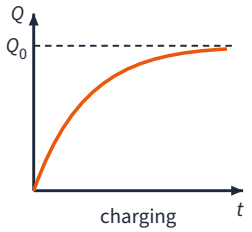
Points must be given in this order. If the capacitor stays connected to the battery, V is constant and Q increases instead.

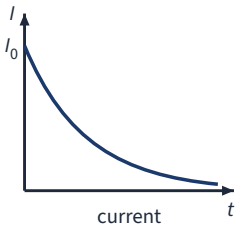
4 CHARGING THROUGH A RESISTOR

SYLLABUS 14.5(B)

RESULT $Q = Q_0(1 - e^{-t/RC})$, $V = V_0(1 - e^{-t/RC})$, $I = I_0e^{-t/RC}$

1. Kirchhoff's law round the circuit (supply V_0): $V_0 = IR + \frac{Q}{C}$, with $I = \frac{dQ}{dt}$.
2. $R\frac{dQ}{dt} = \frac{CV_0 - Q}{C}$, so $\frac{dQ}{CV_0 - Q} = \frac{dt}{RC}$.
3. Integrate from $Q = 0$ at $t = 0$: $-\ln\frac{CV_0 - Q}{CV_0} = \frac{t}{RC}$.
4. $\mathbf{Q = Q_0(1 - e^{-t/RC})}$, with $Q_0 = CV_0$; dividing by C: $\mathbf{V = V_0(1 - e^{-t/RC})}$.
5. $I = \frac{dQ}{dt} = \frac{Q_0}{RC}e^{-t/RC}$: $\mathbf{I = I_0e^{-t/RC}}$, with $I_0 = \frac{V_0}{R}$.





5 DISCHARGING THROUGH A RESISTOR

SYLLABUS 14.5(D)

RESULT $Q = Q_0e^{-t/RC}$, $V = V_0e^{-t/RC}$, $I = I_0e^{-t/RC}$

1. With the supply removed: $0 = IR + \frac{Q}{C}$, and $I = \frac{dQ}{dt}$.
2. $\frac{dQ}{dt} = -\frac{Q}{RC}$, so $\frac{dQ}{Q} = -\frac{dt}{RC}$.
3. Integrate from Q_0 at $t = 0$: $\ln\frac{Q}{Q_0} = -\frac{t}{RC}$, so $\mathbf{Q = Q_0e^{-t/RC}}$ and $\mathbf{V = V_0e^{-t/RC}}$.
4. Size of the current: $I = Q_0e^{-t/RC}/RC$, i.e. $\mathbf{I = I_0e^{-t/RC}}$ (in the opposite direction to the charging current).